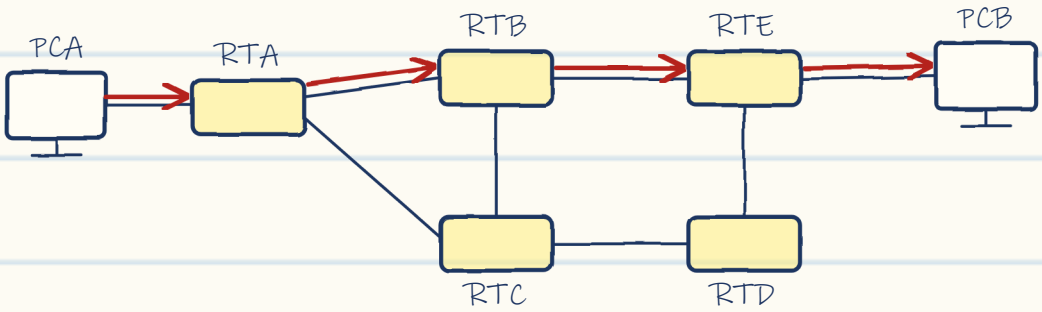
Routerlar, kommutatorlar va ularning OS

Routers, Switches and Their Operating Systems

Asosiy atamalar
- Router — tarmoqlar orasida paketni uzatadi (3-qatlam)
- Kommutator — MAC bo'yicha kadr uzatadi (2-qatlam)
- Marshrut jadvali — maqsad tarmoq, next hop, interfeys
- Comware — H3C qurilmalarining operatsion tizimi

1. Router paketni qanday uzatadi

- PCA maqsad IP ni ko'radi → uzoq tarmoq → default gateway RTA
- RTA marshrut jadvalidan chiquvchi interfeys va next hop ni topadi
- RTE uchun tarmoq bevosita ulangan — paket PCB ga beriladi



RTA marshrut jadvali

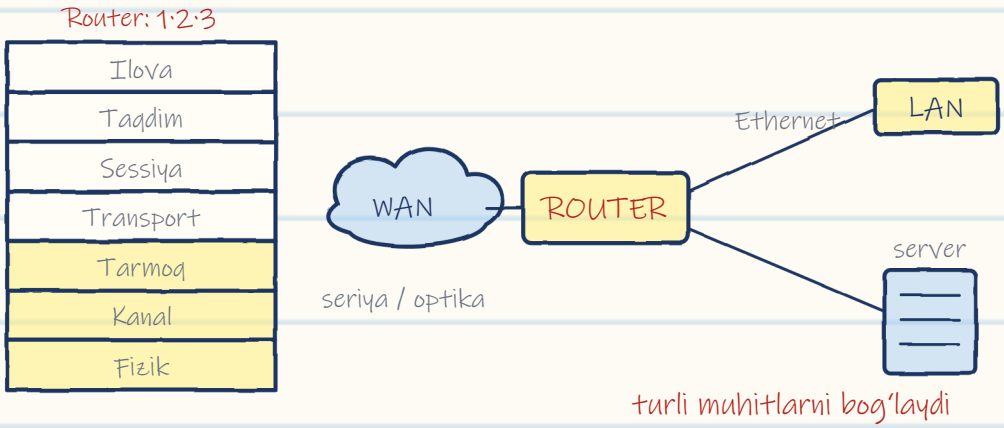
maqsad	next hop	interfeys
10.2.0.0/24	RTB	G0/1
0.0.0.0/0	RTC	G0/2

1-rasm. Har router mustaqil qaror qiladi — jadval bo'yicha keyingi hopga

Yana: router turli muhitni bog'laydi, broadcastni to'xtatadi, marshrutlarni almashib eng yaxshisini tanlaydi.

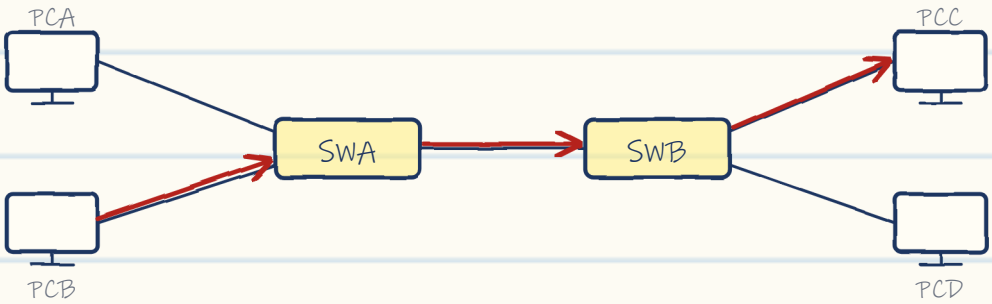
2. Router xususiyatlari

- OSI 1 · 2 · 3 qatlamda ishlaydi
- qaror tarmoq qatlami manzili (IP) bo'yicha
- turli turdagi interfeyslar
- ko'p kanal protokollari
- ko'p marshrutlash protokollari



2-rasm. Bir uchida seriya liniya, ikkinchisida Ethernet

3. Kommutator qanday ishlaydi



SWA — MAC jadvali (o'zi o'rganadi)

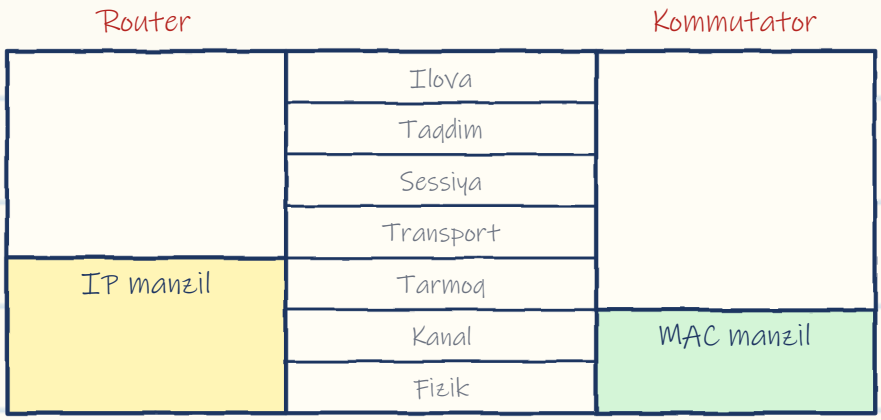
MAC manzil	port
00e0-fc00-1111	G1/0/1
00e0-fc00-2222	G1/0/2

mos keldi → o'sha port; mos kelmadi → hamma portga flood

3-rasm. PCB → PCC: kadr MAC manzil bo'yicha uzatiladi, jarayon shaffof

MAC jadvali: yozuvlar avtomatik o'rganiladi — kommutatorlar bir-biriga hech narsa uzatmaydi.

4. Router va kommutator

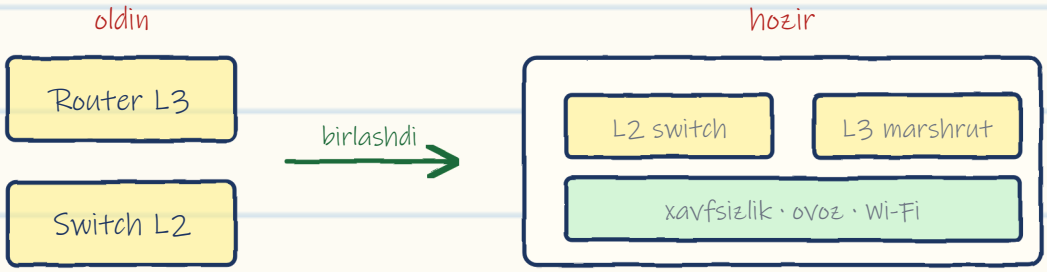


L3 — router · L2 — kommutator

4-rasm. Kommutator 3-qatlam manzilini ham, marshrutni ham ko'rmaydi

	Router	Kommutator
Manzil	IP (L3)	MAC (L2)
Jadval	marshrut — almashinuv	MAC — o'z-o'zidan
Muhit	har xil	faqat Ethernet

5. Rivojlanish trendi

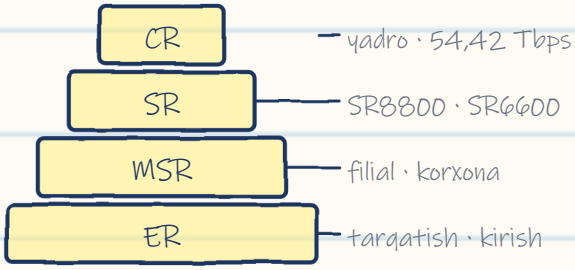


funksiya va xizmatlar birlashdi

5-rasm. Turkumlash: yuqori/o'rta/quyi daraja; yadro · tarqatish · kirish

6. H3C qurilmalari

H3C routerlar



Kommutatorlar

yadro: S12500 · S10500
S9500E · S7500E

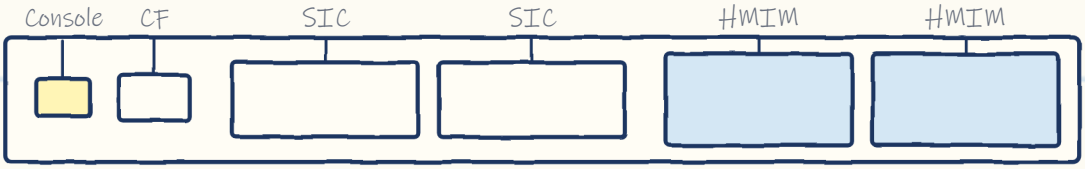
o'rta va kirish:
S5830 · S5820V2 · S5500
S3600V2 · S3100V2

kursda: MSR 36 va S5820V2

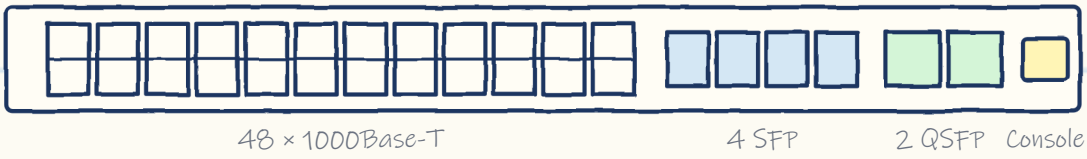
S5820V2-54QS-GE: 48 GE + 4 SFP + 2 QSFP

6-rasm. CR — magistral, SR — yuqori daraja, MSR — filial, ER — kirish

MSR 36-40 — old panel



S5820V2-54QS-GE

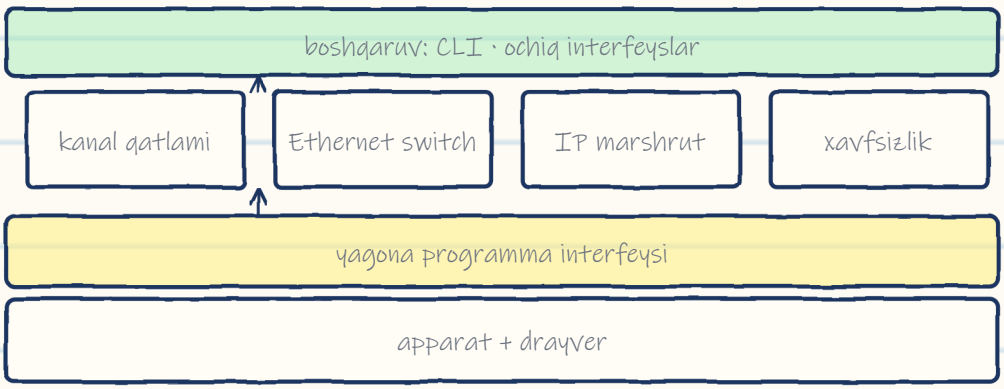


7-rasm. Modulli slotlar: SIC va HMIM; quvvat bloki orqa panelda

> Nomdan o'qiladi: port soni va turi model nomida ko'rinadi.

7. Comware operatsion tizimi

H3C qurilmalarining yagona tarmoq OSi; kursda Comware V5.



8-rasm. Drayver va fizik qism yuqori modullar uchun shaffof

- IPv4 / IPv6 ikki stek
- ko'p yadroli ishlov
- marshrutlash va kommutatsiya — bitta platforma
- yuqori tayyorlik: IRF
- modulli — kengaytiriladi va sozlanadi

Xulosa

- Router — L3 IP manzil, marshrut jadvali, turli muhit
- Kommutator — L2 MAC manzil, jadval o'z-o'zidan, shaffof
- Mos yozuv yo'q → kadr hamma portga flood qilinadi
- Trend: L2 va L3 hamda xizmatlar birlashmoqda
- Comware — H3C qurilmalarining yadro dasturi

Tekshirish

1. Router qaysi OSI qatlamlarida? — 1, 2, 3
2. Kommutator jadvali qanday to'ladi? — avtomatik o'rganadi
3. CR16K kommutatsiya sig'imi? — 54,42 Tbit/s
4. MSR36 modul turlari? — SIC va HMIM